

Homework 2

1. Consider the finite horizon optimal growth model considered in class with log utility and linear production technology.
 - (a) Illustrate on a graph (schematically, no need to use any plotting software) how optimal consumption path evolves over time. What model's parameters play key role in shaping the consumption profile? Interpret intuitively.
 - (b) How does the optimal consumption profile adjust if T increases? R increases? β increases? (illustrate on the graph) Interpret intuitively.
 - (c) What can you say about the properties of the optimal capital profile? If you cannot provide a formal derivation, think intuitively how capital should evolve over time and what parameters impact the properties of the optimal path of capital (in particular, its monotonicity).
2. Consider a *finite* horizon optimal growth model. Suppose that $u(c) = \log c$ and $f(k) = Ak^\alpha$, where $\alpha \in (0, 1)$.
 - (a) (20 pts) Using the Euler equation and the terminal condition, solve for the optimal paths of saving rate, consumption and capital.
 - (b) (20 pts) Find the limit of the optimal path of capital $\{k_t^*\}_{t=0}^T$ as $T \rightarrow +\infty$. Show that this limiting sequence $\{k_t^*\}_{t=0}^{+\infty}$ solves the infinite horizon optimal growth problem (*i.e.*, it satisfies all the sufficient conditions).